I Most Common Vector Fields

Go to goal / Attract

$g = (q_{goal} - q)$

↳ far away you get a large input command

A better design

$\cancel{g \sim \frac{q_{goal} - q}{\|q_{goal} - q\|}}$

$\cancel{g \sim g}$

$g \sim \frac{q_{goal} - q}{\|q_{goal} - q\|}$

Avoid / Repulse

$g = -(q_{obs} - q)$

$g \sim \frac{-q_{obs} - q}{\|q_{obs} - q\|}$

II Shaping Functions - Exponential

A) Exponential

e^{-d^2/σ^2} where d = distance $= \|q_{obs} - q\|$

σ - Width parameter

Limit Influence

Use: Avoid an obstacle

$1 - e^{\frac{-d^2}{\sigma^2}}$

Smooth Convergence

↳ Influence never goes to 0

Not exact for threshold

B) Piecewise

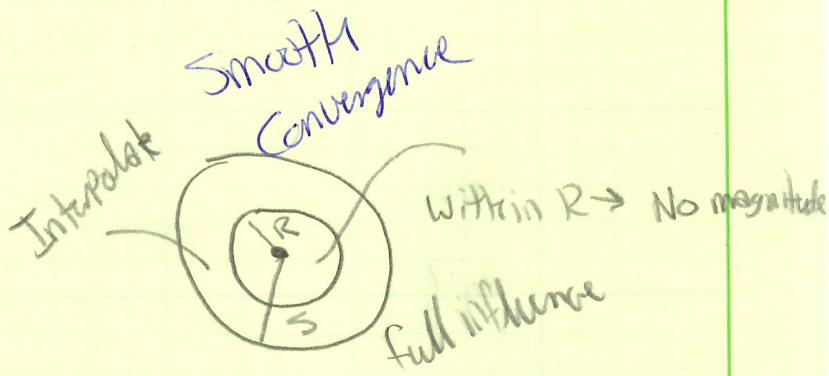
$$\begin{cases} 0 & d > 0 \\ s - d & R < d \leq s \\ 1 & d \leq R \end{cases}$$

s - Sphere of influence

R - Radius of effectiveness

Limit Influence

$$\begin{cases} 1 & d > S \\ 1 - \frac{S-d}{S-R} & R < d \le S \\ 0 & d \le R \end{cases}$$



c) Sigmoid

1) Smooth

$$1 - \frac{2}{1+e^{-d/\sigma^2}}$$

Smooth transition from $1 \leftrightarrow -1$

2) Signed d

$$\begin{cases} 1 & d < 0 \\ 0 & d = 0 \\ -1 & d > 0 \end{cases}$$

III Go-to-Goal

$$g = \frac{v_d}{d}\left(1 - e^{-d^2/\sigma^2}\right)(q_d - q)$$

IV $g = \frac{v_d}{d}\,\text{scale}(d)\,(q - q_0)$ $\quad \text{scale}(d) = \begin{cases} 0 & d > S \\ \frac{S-d}{S-R} & R < d \le S \\ 1 & d \le S \end{cases}$